Rosen — §5.2: Strong Induction and Well-Ordering

Selected Exercises

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Exercise 3

Let $P(n)$ be the statement that a postage of n cents can be formed using just 3-cent stamps and 5-cent stamps. The parts of this exercise outline a strong induction proof that $P(n)$ is true for $n \geq 8$.

(a) Show that the statements $P(8)$, $P(9)$, and $P(10)$ are true, completing the basis step of the proof.

(b) What is the inductive hypothesis of the proof?

(c) What do you need to prove in the inductive step?

(d) Complete the inductive step for $k \geq 10$.

(e) Explain why these steps show that this statement is true whenever $n \geq 8$.

Exercise 9

Use strong induction to prove that $\sqrt{2}$ is irrational. [*Hint*: Let $P(n)$ be the statement that $\sqrt{2} \neq n/b$ for any positive integer b.]

Exercise 11

Consider this variation of the game of Nim. The game begins with n matches. Two players take turns removing matches, one, two, or three at a time. The player removing the last match loses. Use strong induction to show that if each player plays the best strategy possible, the first player wins if $n = 4j$, $4j + 2$, or $4j + 3$ for some nonnegative integer j and the second player wins in the remaining case when $n = 4j + 1$ for some nonnegative integer j.

Exercise 12

Use strong induction to show that every positive integer n can be written as a sum of distinct powers of two, that is, as a sum of a subset of the integers $2^0 = 1$, $2^1 = 2$, $2^2 = 4$, and so on. [*Hint*: For the inductive step, separately consider the case where $k + 1$ is even and where it is odd. When it is even, note that $(k + 1)/2$ is an integer.]

Exercise 13

A jigsaw puzzle is put together by successively joining pieces that fit together into blocks. A move is made each time a piece is added to a block, or when two blocks are joined. Use strong induction to prove that no matter how the moves are carried out, exactly $n - 1$ moves are required to assemble a puzzle with n pieces.

Exercise 15

Prove that the first player has a winning strategy for the game of Chomp, introduced in Example 12 in Section 1.8, if the initial board is square. [*Hint*: Use strong induction to show that this strategy works. For the first move, the first player chomps all cookies except those in the left and top edges. On subsequent moves, after the second player has chomped cookies on either the top or left edge, the first player chomps cookies in the same relative positions in the left or top edge, respectively.]

Exercise 17

Use strong induction to show that if a simple polygon with at least four sides is triangulated, then at least two of the triangles in the triangulation have two sides that border the exterior of the polygon.

Exercise 19

Pick's theorem says that the area of a simple polygon P in the plane with vertices that are all lattice points (that is, points with integer coordinates) equals $I(P) + B(P)/2 - 1$, where $I(P)$ and $B(P)$ are the number of lattice points in the interior of P and on the boundary of P, respectively. Use strong induction on the number of vertices of P to prove Pick's theorem. [*Hint*: For the basis step, first prove the theorem for rectangles, then for right triangles, and finally for all triangles by noting that the area of a triangle is the area of a larger rectangle containing it with the areas of at most three triangles subtracted. For the inductive step, take advantage of Lemma 1.]

Exercise 24

Recall: a stable assignment pairs each suitor with a suitee so that no suitor and suitee who are not paired with each other both prefer each other to their current partners (as defined in the preamble to Exercise 60 in Section 3.1).

A stable assignment is called *optimal for suitors* if no stable assignment exists in which a suitor is paired with a suitee whom this suitor prefers to the person to whom this suitor is paired in this stable assignment. Use strong induction to show that the deferred acceptance algorithm produces a stable assignment that is optimal for suitors.

Exercise 27

Show that if the statement $P(n)$ is true for infinitely many positive integers n and $P(n+1) \to P(n)$ is true for all positive integers n, then $P(n)$ is true for all positive integers n.

Exercise 36

The well-ordering property can be used to show that there is a unique greatest common divisor of two positive integers. Let a and b be positive integers, and let S be the set of positive integers of the form $as + bt$, where s and t are integers.

(a) Show that S is nonempty.

(b) Use the well-ordering property to show that S has a smallest element c.

(c) Show that if d is a common divisor of a and b, then d is a divisor of c.

(d) Show that $c \mid a$ and $c \mid b$. [*Hint*: First, assume that $c \nmid a$. Then $a = qc + r$, where $0 < r < c$. Show that $r \in S$, contradicting the choice of c.]

(e) Conclude from (c) and (d) that the greatest common divisor of a and b exists. Finish the proof by showing that this greatest common divisor is unique.

Exercise 39

Can you use the well-ordering property to prove the statement: "Every positive integer can be described using no more than fifteen English words"? Assume the words come from a particular dictionary of English. [*Hint*: Suppose that there are positive integers that cannot be described using no more than fifteen English words. By well-ordering, the smallest positive integer that cannot be described using no more than fifteen English words would then exist.]

Exercise 40

Use the well-ordering property to show that if x and y are real numbers with $x < y$, then there is a rational number r with $x < r < y$. [*Hint*: Use the Archimedean property to find a positive integer A with $A > 1/(y - x)$. Then show that there is a rational number r with denominator A strictly between x and y.]

Exercise 42

Show that the principle of mathematical induction and strong induction are equivalent; that is, each can be shown to be valid from the other.